claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_f10e69b6-333\agent-a16f05c6248df5c69.jsonl`
- SHA-256: `a65d1ce14cfd7c6ff00e22aa0a4ef2df158847ae5d559cac1321512f5702431c`
- Source modified: `2026-06-15T14:59:15+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_f10e69b6-333`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-15T14:57:50.102Z)

```
You are reviewing a DRAFT product-overview document for "KhelSutra" ? a local, AI badminton highlight/rally tool.
The draft lives at C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md (Read it). It is dual-audience: (a) a non-technical FIELD ASSOCIATE who will talk to
academies, and (b) invited ALPHA/BETA academies/coaches/players who will use the tool and give feedback. It must be
PRODUCT-LEVEL (no fine-grained technical detail) and HONEST (no overpromising to real prospective customers).

GROUND TRUTH (from the repo's docs/memory ? what is actually real):
- TODAY the tool genuinely does: upload a long video ? automatic rally detection ? highlight reel (with a
  "Generated by KhelSutra.guru" watermark) ? a simple summary (rally count + total play time); re-process/download;
  private local-first web app (async jobs + chunked upload + highlight download are live & E2E-verified); runs on real
  4K GoPro match footage.
- Quality reality (DO NOT put raw numbers in a customer doc): clean single-court footage works very well (~0.93 F1);
  busy MULTI-COURT footage is markedly harder (~0.66) and is the explicit moat/focus. The doc should convey this
  qualitatively ("very good on clean single-court; multi-court is harder and is what we're improving").
- ROADMAP (NOT today): shot detection (smash/drop/serve), deeper per-rally/per-match analysis, per-player history,
  "AI assistant coach", more sports (table tennis next), Kannada/Hindi localization.
- THE FLYWHEEL (the headline the owner wants foregrounded): players/coaches can run their OWN videos; coaches can
  correct what it found; over time + more videos the model sharpens AND that library unlocks the deeper analysis.
- GUARDRAILS (hard, non-negotiable, must be represented accurately): using a user's video to improve/train the model
  is OPT-IN (explicit permission), never automatic; footage involving MINORS is NEVER used for model improvement;
  private/local-first ("your video, your data"); pricing is NOT set.
- Capture quality is the gate: fixed camera behind court, full court in frame, landscape, steady, decent light;
  handheld/panning/broadcast footage is the failure mode.
- It is NOT line-calling/scoring/umpiring, NOT player biometrics, NOT a coach replacement.
RULES: read-only; you cannot edit. Return STRUCTURED output (consumed programmatically). Quote the draft line/phrase
when flagging. Be specific and actionable. Do not invent product capabilities.
```

LENS: CLARITY for a non-technical Bengaluru academy reader (coach/owner) AND usability as a ramp-up doc for the associate. Flag jargon, abstractions, or any place a coach would say "so what does it actually do for me?". Check the structure flows for a first-time reader, the length is right for a ramp-up (not too long), the FAQ actually answers what academies ask, and an alpha USER could read it and know how to use the tool well + what feedback to give. Suggest trims of anything that doesn't serve either audience.

2. assistant (2026-06-15T14:57:53.160Z)

I'll start by reading the draft document, then check it against the ground truth from the repo's docs and memory.

3. user (2026-06-15T14:57:53.993Z)

```
1	# KhelSutra ? Product Overview & Early-User Guide
2	
3	*What it does today, how to get the best out of it, and where it's going.*
4	*For field associates and for invited alpha / beta academies, coaches, and players.*
5	
6	---
7	
8	## In one line
9	
10	**KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Point a phone or
11	GoPro at the court, record a session, and KhelSutra finds every rally, stitches a **highlight reel**,
12	and gives you a **simple summary** (how many rallies, how much real play time). No special hardware,
13	no editing, no manual scrubbing.
14	
15	---
16	
17	## The problem we're solving
18	
19	Academies and serious players already record a lot of video ? and then almost never watch it. Hours
20	of footage pile up on phones and memory cards because cutting it into useful clips by hand is tedious
21	and nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22	
23	And most academy halls run **several courts at once**, so a single recording is a busy, noisy scene.
24	Ordinary video tools choke on that. **Pulling the real rallies out of that footage ? automatically ?
25	is exactly what KhelSutra is built to do.**
26	
27	---
28	
29	## Who it's for
30	
31	- **Coaching academies & coaches** ? turn raw session footage into review-ready clips and reels for
32	  teaching, without an editor.
33	- **Serious players** ? get clips of your own rallies to review, keep, and share.
34	- Today's focus is **badminton academies in Bengaluru** ? that's who we're building with first.
35	
36	---
37	
38	## What it does **today**
39	
```

This is what's working right now, in the private web app invited users can try:

- **Upload a full session or match video** (from a phone on a tripod, or a GoPro). Long videos are fine.
- **Automatic rally detection** ? it finds where each rally starts and ends, on its own.
- **Highlight reel** ? the rallies stitched into one clip you can watch and share, with a small
  "Generated by KhelSutra.guru" mark.
- **A simple summary** ? how many rallies were found and how much actual play time there was.
- **Re-use & download** ? process once, come back to it, download the reel.

That's the core loop: **record ? upload ? get back a reel + summary.** No editing skill needed.

How to get the best out of it (capture matters)

KhelSutra works best on a **steady, full-court recording** ? the same way a coach would set a camera to review play. A few simple habits make a big difference to the results:

- **Put the camera behind the court, slightly elevated**, with the **whole court in frame**.
- **Keep it still** ? a fixed tripod beats a handheld or panning shot every time.
- **Shoot landscape**, with reasonable lighting.
- **Record the whole session** and upload it as-is ? KhelSutra does the cutting.

Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style cuts, very dark halls) is the **hardest case** and where results suffer most. When in doubt: fixed camera, full court, don't touch it.

> A one-page capture checklist is provided to academies in the trial.

The part that makes KhelSutra different: **it learns from your videos**

This is the heart of the product, and what sets it apart from a generic "AI clipper":

Your videos make _your_ tool sharper. As you and your players run more of your own footage through KhelSutra ? and as coaches confirm or correct what it found ? the underlying model gets better at *your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything else. It improves with use instead of staying frozen.

That same growing library of real, labelled footage is also what unlocks the **deeper features below** ? which is why early users aren't just customers, they're co-builders. Every session you contribute makes the next version smarter.

Your data stays yours. Using your video to improve the model is **opt-in** ? nothing is used to train anything without your explicit permission. We don't publish anyone's footage. **Sessions involving minors are never used for model improvement.** The product is built **private and local-first**: your video, your data, your control.

Where it's going ? toward an **AI assistant coach**

Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video library above ?
turns it into a genuine training companion:

- **Shot detection** ? recognising smashes, drops, net shots, serves, and clears within each rally.
- **Deeper per-rally and per-match analysis** ? patterns, tendencies, what's working and what isn't.
- **Per-player & per-match history** ? track a player's progress over weeks and months, not just one clip.
- **Personalised insights** ? an **AI assistant coach** that supports (never replaces) the real coach,
  surfacing what to work on next.
- **More sports** ? table tennis next, then other racket/court sports.
- **Your language** ? Kannada and Hindi, alongside English.

> **Read this honestly:** the items in this section are the **direction**, not today's buttons. What you
> can use *today* is rally detection + highlight reels + the play-time summary. The deeper analysis arrives
> as the video library grows ? which is exactly why your footage and feedback matter so much.

What we're focused on over the next ~4 weeks

1. **Growing the library of real academy footage** ? especially busy **multi-court** halls (the hard,
   valuable case). This is the single biggest lever on quality.
2. **Field conversations with academies** ? understanding how coaches and players actually work with
   video, and what would make this genuinely useful.
3. **Recording sessions and showing academies their own footage** as reels ? *"this is what it looks
   like for you."*
4. **Sharpening multi-court accuracy** and inviting a few more academies into the private trial.

What KhelSutra is **not** (so expectations stay honest)

- It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough edges.
- It is **not a line-calling, scoring, or umpiring system** ? it finds rallies and highlights.
- **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available today.**
- It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds insight.
- Accuracy depends on **capture quality**; messy handheld/broadcast footage is a known weak spot.
- **Pricing isn't set** ? we're still learning what's fair; today's invited use is about getting it right.

Quick answers to common questions

"How accurate is it?"

Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what
we're improving. The best answer is to *show* a sample reel rather than quote a number.

"Do I need a special camera or hardware?"
No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page checklist.

"Is my video private? What happens to it?"
It's built private and local-first ? your video stays yours. We never use anyone's footage to improve the
model without explicit opt-in, we don't publish footage, and **footage involving minors is never used**.

"Can it tell me the score / who won / call lines?"
Not today. Today it finds rallies and builds a highlight reel + a play-time summary.

"Does it work for doubles and tournaments?"
Yes for finding rallies. Crowded multi-court tournament halls are the hard case we're specifically working on.

"Will it get better?"
Yes ? that's the whole idea. The more real footage runs through it (with permission) and the more coaches
correct it, the sharper it gets, and the sooner shot detection and deeper analysis arrive.

"Which sports?"
Badminton today; table tennis and other racket sports are on the roadmap.

"When can I use it / how do I join?"
It's an invite-only private trial. If you're interested, the best next step is to let us record a session
and show you your own court and players as a reel.

KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.
Field associates: the conversation script and visit rules live in the separate Field Visit Playbook.

4. assistant (2026-06-15T14:57:56.868Z)

Let me cross-check the draft against the ground truth in the repo's docs and memory to verify accuracy of specific claims.

5. user (2026-06-15T14:57:58.293Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-450 of 1931 total (91083 tokens, cap 25000). Call Read with offset=451 limit=450 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

---
name: current-state
description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
metadata:
  node_type: memory
  type: project
  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301

Current state ? updated continuously (see [[working-agreement-checkpoints]])

Checkpoint: 2026-06-15 (PRODUCT RAMP-UP DOC for the field associate + alpha/beta users). **Owner found a field associate** (willing to talk to coaches/serious players); associate's feedback = he needs to understand the WHOLE TOOL from a PRODUCT perspective before going out. Owner refined scope twice: (1) the doc should ALSO serve invited alpha/beta academies so they can use the tool well + give feedback; (2) **headline the flywheel** ? players/coaches use THEIR OWN videos to fine-tune the underlying model; over time + more videos it sharpens AND grows into shot detection + deeper skill analysis = an **AI Coach**. Building `docs/KHELSUTRA_PRODUCT_OVERVIEW.md` (committable source) + a shareable PDF, PRODUCT-LEVEL only (NO fine-grained tech: no model names/F1/pipeline internals). Honesty rails: clearly separate TODAY (rally detect + highlight reel + play-time summary, phone-on-tripod, private web app) from ROADMAP (auto per-user sharpening, shot detection, AI-coach analytics, multisport, Kannada); consent/opt-in + minors-never-used + local-first must be crisp. Complements (does NOT duplicate) the existing `PRODUCT_FIELD_ASSOCIATE_FLYER.md` (recruiting) + `FIELD_VISIT_PLAYBOOK.md` (visit script). Both repos synced to remote HEAD at session start (indexer 5fe9beb/#176, collector 031e410/#29) ? clean.

Checkpoint: 2026-06-15 (ROORA JUNE-15 BATCH + 3 collector PRs merged + guidelines enriched). Owner-authorized merging of "process" PRs (prep?verify?transcode?pre-ingest?ingest); billing outage still red CI account-wide ? `--admin` merged on local-green (precedent: indexer NEXT_STEPS). **Collector master `031e410`:**
- **#27 preflight gate** (`src/core/batch_preflight.py`): prep-annotation-batch now scans the tree FIRST ? BLOCKS on duplicate videos (size+head+tail fingerprint, never full-hashes 10GB over USB), surfaces empty dirs / VFR / multi-video dirs; `--dry-run` (scan only) + `--allow-duplicates` (for the A/A test). Header-only `proxy.probe_light` (no `-count_packets`) ? 3.8s scan on the real 115GB tree. Caught the empty-TT-dir + Video6/9-byte-dup live.
- **#28 `--encoder nvenc`** (proxy + batch): GPU h264_nvenc (`-rc vbr -cq <crf> -b:v 0`, preset p5); libx264 still default. Parity is codec-agnostic. encoder_available() guards a missing GPU. **NVENC litmus VERIFIED** (RTX 2070S, 36901 frames @ 59.94 both); laptop stays cool (GPU encode, I/O-bound). cq19 ? 2× libx264 size (raise --crf to shrink).
- **#29 richer schema** (court+confidence+capture-report+IAA ? owner picked ALL 4): per-rally `court` (primary|secondary, the multi-court gold label for the 0.66 weakness) + `confidence` (sure|unsure ? `annotator_unsure` review flag, never penalised) carried Brief+Guide+convert-cvat?canonical CSV?round-trip; capture-report (per-video quality form, doc-only) + IAA ~15% A/A overlap (Brief §4/§9). ? court/confidence currently STOP at the canonical CSV+review sheet ? **DB/export integration is a tracked FOLLOW-UP before training on them** (import-local reads core cols only).
- **THE BATCH:** `prep-annotation-batch --encoder nvenc` running on F:\GoPro?\Roora June 15 (12 unique CFR videos, ~115GB 4K/5.3K/one 120fps/one 1080p) ? mirrored proxies-only tree at `G:\My Drive\Sharing\Annotation Videos\Roora Request - June 15` (Google Drive ? shareable). Manifest (private) on F:\?\roora_june15_proxy_manifest.csv; batch log F:\?\roora_batch_log.txt (? python stdout block-buffers ? watch G: proxy files + manifest rows, NOT the log's print lines). ~3/12 done at checkpoint; ~3.7GB/proxy at cq19 ? ~30-40GB Drive upload

(iterate to higher --crf if slow).
- **Roora guidelines Doc** (canonical Google Doc, see [[roora-guidelines-drive-doc]]) now NEEDS regeneration with the new schema ? I can't edit the Doc in place (no Drive write tool); gave owner paste-ready deltas. ?? owner: fill the Doc TODOs + paste the new sections; A/A duplicate-annotation test deferred (owner idea: 2 identical proxies ? check byte-identical CVAT return). DB integration of court/confidence = next collector PR when we train on them.

Checkpoint: 2026-06-14 (? GATE-A REGRESSION RESOLVED + eval?serve FOUNDATION FIXED ? the "owner decision pending" in the checkpoint below is now DONE). The served Gate-A regression (#146 flip ? ?0.008 served, drops ~12 real rallies) is **REVERTED** (**#151** ? served default back to `gemini`, `min_crossings=0`, verified) + the served litmus (**#148**) + finding docs (**#152**) merged. The **foundational eval?serve gap is FIXED** (**#154**): `backend/eval/windowing.py` single-sources the SERVED preset from production `IndexingConfig` (eval+serve can't drift); `backend/eval/promotion.py::promote_if_served_safe` VETOES any cue winning on PROPOSAL windowing but regressing SERVED (Gate-A: proposal +0.0739/6-0 ? served ?0.0077/2-2 ? **promote=False**); `serve_contrast.py` = the worked example. 21 tests, CI green. **[[decision-rigor-norm]] updated with EVAL?SERVE** (a served check is REQUIRED before flipping any served default). Owner directive "do not regress + fix foundations before any other task" ? **BOTH DONE.** ?? **OPEN/HELD:** **#149** (exec-policy P3a worker self-scheduling loop, green, additive) is now UN-BLOCKED but un-merged pending owner go; **#150** (per-user store/bridge v4/v5 + `serving.py`) MERGED additive/inert (NEXT owner-gated = wire the bridge into `_select_for_handoff`); owned-model Phase-0 PLAN at `scratch/OWNED_MODEL_PHASE0_PLAN.md` (item 7, for discussion). Repo clean on `master` ?`4251c9c`; worktree `.claude/worktrees/bhi-execpolicy` remains for #149. **Context ~critical ? keep handoffs granular.**

Checkpoint: 2026-06-14 (SERVED-SIDE GATE-A CHECK ? the #146 flip's offline win does NOT transfer to the production windowing; mild served regression found, owner decision pending). Ran the owner-gated follow-up flagged in PR #146: drove the **live `FusionSegmenter` served seams** (`_enrich_candidate_stats`?net_crossings, `_select_for_handoff`?the gate) over each golden video's **cached** WASB trajectory at the **production windowing** `process_video` actually uses (vt=0.02, merge_gap=2.0, min_win=2.0 from `load_config()`), no WASB re-run / GPU / Gemini. Repro `scratch/served_gateA_{verify,sweep}.py`; artifacts `output/gateA_served_verify/`. **(a) CONFIRMED:** at `min_crossings=0`, fusion_hybrid seeding+handoff is **byte-identical to wasb_hybrid on all 6** (no-regression invariant verified on live served code, not just review). **(b) FINDING:** the +0.074 win was measured on the eval harness's recall-oriented *proposal* windowing (0.005,1.0,0.5; CONTROL over-segments seg-ratio 1.68?1.01 = Gate-A cleaning fragmentation). The **served coarse windowing already merges aggressively** ? Gate-A has little FP to remove and instead trims **real rallies** with net_crossings 0?1. Candidate-handoff F1 (IoU0.5, n=6) sweep: **mc0 0.300 | mc1 0.303 (+0.003, 1 real-rally lost) | mc2(SHIPPED) 0.292 (?0.008, 12 real-rallies lost) | mc3 0.269 (?0.031, 28 lost)**; worst largetest_doubles ?0.051. Tiny/within-noise BUT mc2 is **not** a served improvement and drops genuine rallies (dropped windows never reach Gemini = unrecoverable recall loss; precision "gain" partly duplicates Gemini ? served tradeoff ? neutral). **Root cause = eval?serve windowing** (harness promotes on a windowing production doesn't serve). **Recorded in `docs/QUALITY_ITERATIONS.md`** ? committed as **PR #152 (merged, `53bc743`)**. **? RESOLVED IN PARALLEL by a SIBLING session:**

the same regression was independently found + productionized as a standing litmus (**#148** `backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`) and the **flip was REVERTED in #151** (`config.json` default_segmenter ? `gemini`, `indexing.fusion` block removed; `FusionConfig.min_crossings` default 2?0 ? Gate A OFF-by-default-but-opt-in). My independent numbers MATCH #151's (?0.008, ~12 rallies dropped, 2/6 losses) = good corroboration. So the owner decision I'd queued is MOOT (already reverted). Reconciled my #152 doc entry to point at the revert (#151) + the productionized guard (#148) instead of "decision pending". **OPEN follow-up (real lesson):** align the experiment-harness windowing with the served windowing before treating a harness lift as a served default. Ties [[decision-rigor-norm]] (foundations: eval?serve) + [[weak-signals-retained]] + [[paper-coauthor-aim]]. ? Watch [[gotcha-shared-checkout-branch-collisions]]: this whole arc (#148/#151 sibling vs my #152) was concurrent slates on the same finding ? branch-from-origin protocol held.

Checkpoint: 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint). **Owner adopted the HYBRID** ? personalization as a flag-gated FEATURE on a generalization-first baseline + a **contribution flywheel** ("a tool that gets better the more you help it"; free-tier videos pool into the owner-trained baseline). Pivot analysis (`scratch/PERSONALIZATION_PIVOT_ANALYSIS.md`) recommended hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-loss objection. **MERGED this session:** telemetry wire-in **#140**; per-user promotion gate **#141** (min-N floor + structural-guard exemption); held-out-USER learning curve **#142** ("<K videos to superior quality" = falsifiable); per-user no-regression guard **#143**; **`docs/PERSONALIZATION_PLAN.md` #144** (the checked-in project doc + locked decisions). Gemini-training-source concern = **CLEAN** (eval-only, gitignored, M0.3 owner-gated). **LOCKED owner decisions:** identity = north-star-design/current-impl (nullable user_id, device UUID); inline classifier_json behind a cloud-load interface; **`min_n` is a PROMOTION-confidence floor, NOT a service gate** (a 1-video user is always fully segmented by the baseline); free-tier video pooling (consent + minors-exclusion); **Gate-A-only structural + `min_n=5`**. ? **GATE-A MILESTONE DONE:** Gate-A FLIPPED to the served default (**#146**, config-reversible ? `default_segmenter` gemini?`fusion_hybrid` + `min_crossings` 0?2; rollback = one flag); the **ablation layer** built + **LITMUS PASSED** (**#147** ? `backend/eval/ablation.py` LOSO/LOGO runner reproduces Gate-A +0.074 / 6-of-6 / p=0.031 EXACTLY; PDF export ? also in `OneDrive\Documents\Rally_Ablation_Report.pdf`); quality-iterations checkpoint (**#145**); **[[decision-rigor-norm]]** codified (NUMBERS INFORM, FOUNDATIONS DECIDE). Numbers: Gate-A = the ONLY lever clearing the honest bar; audio +0.079 = suggestive lead (CI spans 0); Gate-B inert/trap (court-mask dep). **GPU + Gemini AUTHORIZED for evals** (cooler boost on). ?? **OPEN:** served `fusion_hybrid` path is **mock-tested-only on real video** ? a **served golden regression** is recommended (re-score cached WASB ? fusion-default vs golden labels); `reportlab`/`pypdf` ? requirements (CI-exercise the PDF). NEXT personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into `_select_for_handoff`) ? owner-gated, see [[weak-signals-retained]] + `docs/PERSONALIZATION_PLAN.md`.

Checkpoint: 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134 merged + Cooler Boost). **THE NUMBERS (honest LOVO n=6, IoU?0.5, C1 bootstrap-CI + sign-test):** shuttle-only F1=**0.307**; **+person (P3) ?F1=?0.021** (?6.7%, CI [?0.165,+0.162], sign 2W/4L p=0.688, **CI-clears-0: NO**) ? NO lift, stays default-OFF (no court polygons ? `players_in_court` counts spectators; per-video wash: testlarge +0.40 / gbaaddy ?0.22). **+person+audio (P4 incremental) ?F1=+0.079** (+27.7%, F1

0.286?**0.366**, CI [?0.054,+0.203], sign 4W/2L p=0.688, **CI-clears-0: NO**) ? PROMISING (wins 4/6, largetest/mahadevpura +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. **VERDICT: neither promotable (both CIs span 0); audio is the lead ? need more matches + measure audio on shuttle-only directly (drop the person handicap).** Recorded in `eval_baselines/experiment_leaderboard.json`; honest negative/uncertain results KEPT ([[paper-coauthor-aim]]). **LANDED THIS SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137); `experiment_leaderboard.json` tracked (#139); **weak-signals audit ?** (person/audio/court-bounds all present, default-OFF, no-special-casing, retained ? refuted=false ? [[weak-signals-retained]]); 7-page paper-style **quality report PDF** at `OneDrive\Documents\Rally_Detection_Quality_Report.pdf` (gen `scratch/quality_report.py`; conservative academic restyle, strict-reviewer-approved); `FusionGoldenExtract` scheduled task **unregistered** (cleanup); tree clean on `master`. ?? **NEXT (fusion track):** (1) **wire `run_telemetry` (#134) into fusion_golden + the ps1 wrapper** (deferred, ready, mergeable); (2) **measure audio on shuttle-only directly** (quick eval, reuses the 6 `output/*_golden_features.json`, NO new data ? audio's standalone lift without the person handicap); (3) **grow the golden corpus via Roora** (owner's job ? gives power to confirm the +0.079 audio lead) + **activate court polygons** (re-measure person properly ? the ?0.021 is partly a `court_polygon=None` artifact); (4) future **shuttle-dropped** cue ([[weak-signals-retained]]). **OTHER track (per [[session-handoff]] ? the stated IMMEDIATE PICKUP): EXECUTION-POLICY P3 then P4** (owner-gated; #132/#133/#135 done). **Extraction infra** (all working): detached Scheduled Task `FusionGoldenExtract` runs `scratch/extract_until_done.ps1` (self-restart loop + in-process keep-awake + GPU-busy gate: waits if free VRAM<3500MB) ? deaths were `0xC000013A` console-control kills (NOT crash/sleep/TDR/OOM ? verified via event log), auto-recover via 3-min repetition trigger + restart-on-failure. **Cooler Boost** (MSI Creator Center) dropped 85?67°C, SM ~1100?~1300MHz, velocity ~26?~50 fps (the rest is Max-Q power cap, not thermal).

Checkpoint: 2026-06-14 (EXECUTION-POLICY framework: design + P1 + P2 SHIPPED ? **PR #132 (design) + #133 (P1) + #135 (P2) MERGED**, self-merged on green CI). Owner asked to harden the GX010125 Gemini hang as a **policy-driven, per-job, self-scheduling** system (default WAIT_AND_RESUME, ABORT, POLL_EVERY_N_SEC?, **no master node**) with **very high coverage** (failures cause delays) and **clean poll-logging** (visible but not polluting). **#132** = design doc `docs/EXECUTION_POLICY.md` (validated the owner's architecture: per-job policy = data on the job; self-schedule via re-enqueue; no master ? rides the existing single-worker executor + `jobs` table + crash sweep; DB is the coordinator). **#133 (P1, the actual fix)** = `backend/infrastructure/execution_policy.py`: `ExecutionPolicy` + `run_bounded` (HARD `op_timeout_sec` per-attempt ? the hang-killer; retries failures AND hangs; hung attempt's daemon thread abandoned) + `poll_until` (bounded poll loop). Clocks injected ? **12 tests, 100% module coverage, instant**. Refactored `gemini.py`: `generate_content`?`run_bounded` (a never-returning generate previously blocked FOREVER = the GX010125 hang; now bounded), processing-wait?`poll_until` (**per-poll logs at DEBUG, not an INFO line every 10s** ? the no-pollution ask). 23 gemini tests stay green. **#135 (P2, substrate)** = DB migration **v3** (`jobs.policy` JSON + `next_poll_at`; existing rows survive via ALTER) + `set_job_waiting(delay_sec)` (self-schedule: park 'waiting' until due) + **`claim_due_job()`** (atomic compare-and-set claim of queued/due-waiting ? running; concurrent-safe, no master) + `create_job(policy=)`/`get_job` policy. **Additive, no executor change**; 10 tests;

145 version asserts bumped 2?3. ? **THIS IS THE OWNER'S REAL GPU MACHINE** (torch+CUDA/cv2/WSL/Gemini live). ?? **REMAINING
146 (owner-gated, NOT done ? deliberately not rushed at session tail given the high-coverage bar): P3** = wire the executor
147 (main.py) to a `claim_due_job` self-scheduling loop honoring WAIT_AND_RESUME (release+reschedule) + **resumable AI handoff**
148 (persist per-window confirmation; resume only unconfirmed ? the GX010125 reel becomes exhaustive across resumes, no wasted
149 Gemini spend); **P4** = apply the policy to the WSL/WASB GPU pass + ffmpeg. Each = its own distinct PR. Ties [[paper-coauthor-aim]] indirectly (infra).
150
151 **Checkpoint:** 2026-06-14 (ACTED ON FIRST-A/B FINDINGS: Gate-A PROMOTED + learned FIXED + GX010125 run ? **PR #130 + #131
152 MERGED** to master `2e5892e?`; self-merged on green CI). ? **THIS IS THE OWNER'S REAL GPU MACHINE** (torch+CUDA, cv2, WSL
153 Ubuntu, Gemini key in .env all present) ? not the dev container the older CLAUDE.md note assumes. **#130 = learned-variant
154 FIX**: additive/default-OFF `Variant.tune_threshold` (operating point on the TRAIN fold) + `calibrate` (Platt/isotonic, C2);
155 validated on n=6 golden ? **mean held-out F1 0.307?0.423 (+0.116), testlarge 0.000?0.300 (un-zeroed), now BEATS CONTROL
156 0.388**; threshold-in-LOVO is the big lever. **#131 = Gate-A PROMOTED** (first cue promoted via the harness; doc trail in
157 QUALITY_ITERATIONS ? measured win dF1 +0.074, 6/6, p=0.031; recommended `indexing.fusion.min_crossings=2`; **live served-default
158 flip stays OWNER-GATED** until a real fusion-path run confirms ? fusion P2 is unverified on real video) **+ new
159 `docs/PER_VIDEO_OUTPUT_LAYOUT.md`** (one dir per video; fixes the hardcoded `"output"` joins in `trajectory_hybrid.py:~176`
160 so `--output-dir` truly isolates). **GX010125_1080p RUN (offline, no GPU):** ? key finding ? `wasb_runner.run_predict` has
161 **NO cache-hit early-return**, so the full pipeline would RE-RUN WASB (~tens of min on 1080p) + overwrite the cached CSV; the
162 cached trajectory is reusable ONLY via offline re-score. So ran `scratch/run_gx010125_rescore.py` = parse cached
163 `output/wasb/GX010125_1080p__89605e80ed01_wasb.csv` ? 45 candidates ? CONTROL 45 rallies (11.0min) vs **Gate-A 27** (9.3min),
164 Gate-A **dropped 18 low-crossing windows (~1.7min)** of likely held-shuttle/feed FPs ? isolated `output/GX010125_run/` (no
165 pollution; no golden labels for this footage ? divergence not F1). A FULL Gemini-confirmed highlight reel = a separate
166 ~tens-of-min GPU+API run (owner kicks off). Repro: `scratch/{run_gx010125_rescore,ab_n6_lovo,ab_eval_demo}.py`. All in
167 isolated worktrees off master (siblings concurrent). ?? NEXT (owner-gated): real fusion-path run to confirm Gate-A served-side
168 + flip the live default; impl `PER_VIDEO_OUTPUT_LAYOUT` L1; Tier-2/3 §13 corrections; analyzers/reconcilers A1?A5/R1?R3. Ties [[paper-coauthor-aim]].
169
170 **Checkpoint:** 2026-06-14 (FIRST GOLDEN A/B EXPERIMENT + HARNESS HONEST-REPORTING WIRED ? **PR #128 + #129 MERGED**
171 to master `f680b8d`, self-merged on green CI per [[working-agreement-merging]]; owner-directed momentum). Ran the
172 **first real end-to-end A/B on the n=6 golden corpus, 100% OFFLINE** (re-scored persisted WASB trajectories via
173 `distill_local.build_candidates`, no GPU ? the "persist once re-score" thesis proven). One candidate set re-scored
174 three ways: **Gate-A (net_crossings?2) is a MEASURED WIN ? ?F1 +0.074, 6/6 matches, sign-test p=0.031, CI
175 [+0.042,+0.104], seg-count-ratio 1.68?1.01** (over-seg nearly eliminated); the **window-level 5-feature logistic did
176 NOT clear the bar** (?0.081, 2W/4L; testlarge zeroed = threshold/calibration artifact ? C2 motivation; **distinct from
177 the stronger per-frame `rally_seq_proto` 0.583** ? don't conflate). Headline: **the framework held up ? C1 promoted one
178 variant and refused another (right call both ways); C4 surfaced over-seg tIoU hid.**

#128 = archived posterity doc
179 `docs/archives/quality-iterations/2026-06-first-ab-experiment/` (full tables) + archives-index + live QUALITY_ITERATIONS
180 pointer. **#129 = (b) wired C1+C4 into the CORE `experiment.compare()`** (additive/default-ON `honest_stats=True`:
181 `honest` block = per-metric bootstrap CIs + paired ?F1 CI + exact sign test + F1@k + segment_count_ratio; `record_experiment`
182 carries honest ?F1; existing fields byte-unchanged, `honest_stats=False` = legacy shape). Reuses #124 helpers; complements
183 sibling **#127** which wired the same C1 into `fusion_compare` (no dup). 19 existing experiment tests pass + 3 new;
184 ruff+mypy+CI all green. Repro `scratch/ab_n6_lovo.py` + `scratch/ab_eval_demo.py` (machine-side; read gitignored
185 manifest/external golden). All done in **isolated worktrees off master** (siblings merged #121-#127 concurrently). ?? NEXT
186 (owner-gated): promote Gate-A via the no-regression gate (it's a real measured win); fix the logistic (C2 calibration +
187 threshold-inside-LOVO); Tier-2/3 §13 corrections; analyzers/reconcilers impl A1?A5/R1?R3 (none started). Ties [[paper-coauthor-aim]].
188
189 **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASUREMENT BUILT + GPU EXTRACTION RUNNING ? master `d7b1f51`; suite 514 green).
190 The "modeling" track for the person-cue (P3) + audio (P4) fusion lift ? all default-OFF, no-special-casing, harness-measured.
191 **Merged this run (mine): #121** fusion P3 (`backend/eval/fusion_golden.py` offline person-feature EXTRACTOR + `fusion_compare.py`
192 LOVO driver ? reuses build_candidates/PersonDetector/fusion_features/experiment.compare, NO new scoring); **#122** P4
193 (`backend/pipeline/detectors/audio_features.py` + `backend/eval/fusion_audio.py` incremental audio cue); **#123** progress-logging
194 + resumability (`--fresh`); **#125** single-pass `PersonDetector.detections_over_windows` (fixed the per-window-seek O(N)
195 pathology ? ~28s/window ? one sequential decode); **#126** `--smallest-first`; **#127** honest ?F1 = bootstrap CI + paired
196 sign test on per-video held-out F1 (wires #124's `eval_stats.paired_delta_ci`; "never a bare mean"). Concurrent sessions
197 merged #119 (analyzers framework), #120 (run.log), #124 (eval-honesty utils: eval_stats/score_calibration/segmentation_metrics).
198 **? GPU EXTRACTION via DETACHED SCHEDULED TASK `FusionGoldenExtract`** (runs `scratch/extract_until_done.ps1` ? Task Scheduler
199 owns it, OUTSIDE Claude's job object ? survives turn/context/session boundaries; in-process keep-awake + self-restart loop until
200 6/6; clean UTF-8 logs to `output/fusion_golden.log` + `scratch/extract_wrapper.log`). Loops `python -u -m backend.eval.fusion_golden
201 --manifest output/human_lovo_manifest.json --out-dir output --smallest-first` ? 6 `output/<name>_golden_features.json` (GITIGNORED
202 + RESUMABLE: skips any video whose JSON exists). **3/6 DONE** (testlarge, mahadevpura, largetest); remaining
203 kushagra?gbaaddy?adarsh(last, 84k frames). **MONITOR:** JSON count + `tail output/fusion_golden.log` + `nvidia-smi`;
204 `Get-ScheduledTask FusionGoldenExtract|Select State`. **RESTART (resumes, skips done):** `Start-ScheduledTask -TaskName
205 FusionGoldenExtract`. **STOP:** `Stop-ScheduledTask -TaskName FusionGoldenExtract` + `Get-Process python|Stop-Process -Force`.
206 ?? **WHY DETACHED (debugged 2026-06-14):** Claude-bg-task runs DIED TWICE silently (largetest 61%, kushagra 6% @137s), NO Python
207 traceback AND ? verified via Windows event log ? NO sleep (Kernel-Power 42), NO TDR (Display 4101), NO WER crash, NO RADAR-OOM
208 (no event 1003; 18GB free). Traceless vanish = external `TerminateProcess`; common factor = both were Claude-managed bg tasks
209 (job-object kill-on-close at a context boundary fits death #1 @ the prior context summary). Scheduled Task = immune; GPU access
210 CONFIRMED in the interactive task session (53?64% util, 3.1GB). GPU util while running = 20?64% bursty ? batch=1 Faster-RCNN
211 latency-bound (small kernels + CPU NMS + per-frame ByteTrack), NORMAL, model IS on CUDA,

NOT decode-bound; batching the forward
212   pass is the next-extraction speed lever (changes features, so NOT mid-run). ~1.5-2h total. Stale Claude bg keep-awakes stopped. **DEATH #3 + GPU-job check (2026-06-14):** died a 3RD time (gbaaddy 69%)
213   ? `LastTaskResult=0xC000013A` = STATUS_CONTROL_C_EXIT = terminated by a **console-control event / CTRL+C** (NOT crash/sleep/TDR/OOM
214   ? explains the no-trace vanish; killed the detached task too, so NOT Claude's job object). GPU-contention angle CHECKED: **NO
215   competing GPU compute job** (only desktop apps + our 1 python; 5GB free; HW is a **laptop** RTX 2070S **Max-Q**, MSI). Mitigations:
216   (1) task re-registered with **3-min repetition trigger + restart-on-failure (count 10, 1-min)** ? auto-recovers within minutes of
217   ANY death; (2) **GPU-busy gate in the ps1** (owner policy = *wait until free, auto-resume*): before each python launch, if free
218   VRAM <3500MB (another job has the GPU) it waits 30s + re-checks, then launches. **?? WHEN IT COMPLETES ? the remaining steps for
219   the NUMBERS:** (1) `python -m backend.eval.fusion_compare --features-dir output --record` ? honest P3 LOVO ?F1 (shuttle-only
220   vs +person) w/ CI+sign-test; (2) `python -m backend.eval.fusion_audio --augment --compare --record` ? P4 incremental audio
221   ?F1; (3) log both to `docs/QUALITY_ITERATIONS.md` I3 (HONEST measured numbers only ? [[paper-coauthor-aim]]). Honest caveat:
222   NO court polygons ? `players_in_court` counts spectators (3-7 on "single-court" Mahadevpura); robust signals = motion +
223   `shuttle_player_colocation`. GPU = RTX 2070S, person detection on Windows torch cu124 (NO WSL). Branch protocol held through
224   heavy concurrent merges (#119-127) ? [[gotcha-shared-checkout-branch-collisions]]. On master, clean.
225   **STORAGE (this session, owner-approved "delete cache + compact WSL"):** deleted the stale **93GB GX010125 frame cache**
226   (WSL `~/clips` 95?2.3GB; WSL fs used 152?60GB, 896GB free; C: 53?72GB free) ? kept the 2.2GB GX010125 proxy. The WSL
227   `ext4.vhdx` was `--set-sparse true` + `fstrim`'d (946GB trimmed) but this WSL build (2.3.24) did NOT auto-deallocate on
228   shutdown ? the file is still **~221GB allocated on C:**. **DEFERRED to post-extraction: the physical ~150GB C: reclaim via
229   `wsl --shutdown` then elevated `diskpart` ? `select vdisk file="...\CanonicalGroupLimited.Ubuntu_79rhkp1fndgsc\LocalState\ext4.vhdx"`
230   ? `compact vdisk`** (heavy C: I/O would fight the GPU run's video reads). [[gotcha-long-runs-gpu-wsl]] · [[cache-hygiene-policy]].
231
232   **Checkpoint:** 2026-06-14 (RALLY-DETECTION FRAMEWORK doc + LITERATURE REVIEW + eval-honesty code ? **PR #119 + #124 MERGED**
233   to master via self-merge on green CI per [[working-agreement-merging]]; owner-directed). **PR #119** (`docs/ANALYZERS_AND_RECONCILERS.md`):
234   reframed the analyzers/reconcilers design into a **multi-scope signal-fusion framework** (frame detectors / window analyzers /
235   **session analyzers**; warm-up `SessionAnalyzer` + shuttle-state frame-detector folded in) + **§13 = a verified multi-agent
236   literature-review** (8 patterns surveyed, ~40 citations adversarially checked, **ZERO fabrications**; author-line fixes applied).
237   **Verdict (honest):** the skeleton is STANDARD prior art ? candidate-windows+scorer = Temporal Action Localization;
238   reconciler-as-refinement = Temporal Action Segmentation post-proc; "signal not verdict" = late/score-level fusion + stacked
239   generalization (~20 yrs old, already in racket sports). **Only the report-first reconciler-as-linter bundle rated
240   `novel-combination`**; warm-up-as-soft-feature = thinnest gap. **Cite, don't claim**; terminology realigned (producer?labeling
241   function, scorer?stacked meta-learner, LOVO?leave-one-group-out, tIoU?F1@k/mAP). Ties [[paper-coauthor-aim]] (§13 is the honest
242   related-work backbone). **PR #124** implements the **Tier-1 measurement-honesty course corrections C1?C4** as PURE ADDITIVE /
243   default-OFF helpers (no existing compare/LOVO/metrics behaviour changed): `backend/eval/eval_stats.py` (C1 `mean_with_ci`/

`bootstrap_ci`/`paired_sign_test`/`paired_delta_ci` ? never a bare LOVO mean at n?6; C3 `out_of_fold_predictions`/`grouped_folds` ? no stacking leakage), `score_calibration.py` (C2 Platt+isotonic+brier/ECE ? comparable cue confidences), `segmentation_metrics.py` (C4 F1@{10,25,50} + mAP@tIoU + segment_count_ratio ? make over-segmentation tIoU is blind to visible). 24 tests; ruff+mypy clean; all 4 CI lanes green (full-suite coverage floor held). Both PRs done in **isolated git worktrees** (off master) to dodge the recurring shared-checkout branch-flips (siblings merged #117 fusion-P2, #118, #120, #123 concurrently; **#116 closed ? its content had landed via #117**). ?? NEXT (owner-gated): **wire C1?C4 into the DEFAULT harness reporting** (`compare`/`run_regression.py` emit CIs + F1@k/edit-score instead of a bare mean ? changes reported numbers, so its own PR) + the Tier-2/3 corrections in §13 (global reconciler decode, cue-correlation modeling, temporal smoothing, external ShuttleSet eval). Analyzers/reconcilers IMPL (A1?A5/R1?R3) still owner-gated, none started.

Checkpoint: 2026-06-14 (RUN.LOG READABILITY for manual QA ? **PR #120 MERGED ? master `6076eef`**; owner asked ? self-merged on verified-green CI per [[working-agreement-merging]] (squash; remote+local branch deleted); was branch `feat/runlog-readability` off `35a50f3`; CODE, minimal/config-gated). Owner's manual-QA loop = run the tool on a video ? eyeball the rally CSV against `run.log`. Two readability changes: (1) **suppress HTTP chatter** ? new central helper `backend/utils/logging_setup.py::setup_logging(verbose_http=False)` raises `httpx/httpcore/urllib3/google_genai/google.genai/google.auth/grpc` to WARNING (NOTSET to restore), gated by NEW knob `logging.verbose_http: false` (`LoggingConfig` on `AppConfig`); our `backend.*` loggers untouched; wired into `run_process_and_stitch.py` (at import + re-applied post-config-load). (2) **end-of-run rally summary** ? new pure helper `backend/tools/export.py::format_rally_summary` (next to `segments_to_csv`/`basic_stats`): header (total rallies + total play HH:MM:SS) + table `# start end dur(s) shot_count` from the produced play_segments so log?CSV line up row-for-row; handles metadata dict-or-JSON-string, missing shot_count?`-`; HH:MM:SS via local `_fmt_hms` matching the segmenters' `_fmt_ts`; logged as ONE block at end of `main()` (also on the all-filtered early-return). Tests: `test_logging_setup.py`, `test_rally_summary.py`, +`verbose_http` default/override in `test_config.py`. Suite **477 green**, ruff (`.`) + mypy (`backend training`) clean. ? Commit-msg first attempt mangled (PowerShell `@'?'@` here-string in the *Bash* tool ? literal `@`; amended via `-F file`) ? Bash tool ? PowerShell. **BRANCH-COLLISION CLEANUP (owner asked):** after merge, LOCAL checkout left clean ? on `master`==`origin/master`==`6076eef`, 0 ahead/behind; pruned 6 stale LOCAL branches mapped to MERGED/CLOSED PRs (#92/#104/#111/#106/#102 + #116-closed-redundant) + the stale remote-tracking refs; **kept** `docs/analyzers-framework` (#119 OPEN). Working tree = only the intentional untracked carryovers (`scratch/`, `artifacts/`). REMOTE still has 15 prunable branches (14 merged-PR + #116) ? **offered to prune, owner did NOT answer the question ? left untouched** (must KEEP `feat/khelacharya-shot-intelligence` #11 = deliberately-retained sibling layer per [[sibling-internship-repo]], + #119). ?? NEXT: owner does the machine-side check (run on a real video ? eyeball the quieter log + summary block; dev container can't run the GPU pipeline); remote-branch prune available on request.

Checkpoint: 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN, owner-gated, do NOT merge**; branch `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas this session and asked to fold them in and treat the doc as a **reusable framework / research paper**, not a project patch (ties [[paper-coauthor-aim]]). Extended the on-master `docs/ANALYZERS_AND_RECONCILERS.md` (which had landed via #117) from "two seams" into a **multi-scope signal-fusion framework**: producers emit SIGNALS at **3 temporal scopes** ? frame detectors (`RallyCue`) / window analyzers / **session analyzers** ? all consumed as feature columns by the learned scorer (spine unchanged: **no special-casing**, signal ?scorer, never `if/else`). Folded in: (1) **warm-up/practice** as a new **`SessionAnalyzer`** tier (whole-timeline span ? an `in_warmup` membership feature the scorer DOWN-WEIGHTS, NOT a hard `if t<warmup_end: drop`; measurable on golden TODAY since warm-up is already labeled negative); (2) **shuttle-state** (`in_air`/`racket_contact`/`inactive`) placed correctly as a **frame-scope detector** (a MULTI_SIGNAL_FUSION_PLAN `RallyCue`, NOT a window analyzer) ? derive-from-existing-signals first (velocity+visibility+direction-reversals+audio thwack), dedicated per-frame model ONLY where WASB trajectory gaps bite, with the per-frame-label blocker flagged. Added **§10 generality & research positioning** (weakly-supervised temporal segmentation; standard-vs-contributed) with an explicit **attribution honesty caveat**: NO fabricated citations/results, related-work + ablations + >1-sport still owed, n=6 = direction-check only ? the honest backbone the paper aim needs. Diff = `2-seam?framework` (310+/137?). ? Shared checkout had flipped to master mid-task (the sibling's collision lesson recurred) ? re-verified branch, branched fresh, normal commit (no in-flight WIP left to isolate; #117/#118 already merged so all cross-links resolve). PR #116 stays CLOSED (content on master via #117). ?? NEXT (owner-gated): review #119; impl tracks unchanged (fusion P3 LOVO, then analyzers/reconcilers A1?A5 / R1?R3 per the doc's phasing ? each ONE PR, default OFF, promote on measured LOVO lift).

Checkpoint: 2026-06-14 (FUSION P2 SHIPPED + collision cleanup + paper aim ? master `35a50f3`; suite 467 green; 0 open PRs). **PR #117 MERGED = `FusionSegmenter`** (fusion P2, registry `fusion_hybrid`, **default-OFF**): shuttle-primary (windows seed from WASB/TrackNet) + adjudicating cues ? `net_crossings` (Gate A, always persisted, GPU-free via rally_gate w/ a reused axis estimate) + optional person cue (Gate B) persisting the ~5 `fusion_features.py` features (pure-tested) via `PersonDetector.detections_per_frame` over each window's ORIGINAL-video frame range. Plugs in via **two IDENTITY hooks** on `trajectory_hybrid` (`_enrich_candidate_stats`/`_select_for_handoff`) ? wasb/tracknet byte-identical (adversarially verified). Served gating optional+default-OFF (`fusion.min_crossings`/`min_players`) ? gates the handoff while persisting the full SUPERSET (offline substrate intact). Court mask = optional `fusion.court_polygon`. **Design-first** (`fusion-p2-design` workflow) ? **31-agent adversarial review** (25 findings?19 confirmed; headline = NO-REGRESSION verified) ? fixes folded (frame-dim guards, round() frame idx, rally_gate `estimate=` reuse, honest torch docstrings, +6 tests). **PR #118 MERGED** = README index fix (QUALITY_ITERATIONS + EXPERIMENT_HARNESS P1-SHIPPED). **? SHARED-CHECKOUT COLLISION (lesson):** the spawn_task DESIGN session ran in THIS checkout, committed `0f673ff` (the analyzers design doc) + flipped the branch BETWEEN my branch-point check and my `checkout -b` ? my fusion branch rode on top ? **#117's squash absorbed the design doc**, so **`docs/ANALYZERS_AND_RECONCILERS.md` is on master via #117** and **PR #116 was CLOSED as redundant** (content intact on master; #116's stale "OPEN do-not-merge" checkpoint below is superseded). Re-verify `git branch` AND that no sibling commit landed since, before `checkout -b`. **Owner side-aim ratified: co-author a paper** "local, cheap & efficient rally detection via multiple clever ideas" ? [[paper-coauthor-aim]]; `docs/QUALITY_ITERATIONS.md` is the ablation-ready empirical backbone (honest measured results only). **Analyzers+reconcilers DESIGN on master (owner-gated, NOT started):** injectable per-window analyzers (e.g. net-hit service-fault) + a report-first reconciliation pass; spine = signals?learned-scorer, **no special-casing** (owner directive). ?? NEXT (owner-gated): **fusion P3** (person features ? `experiment.compare` LOVO lift on the 6 golden videos ? the eager-person-compute cleanup lands here) + measure the Gate-A held-shuttle drop on GPU; then analyzers/reconcilers impl; P4 audio. Earlier this session: PR #114 harness, #115 person-cue extraction (see checkpoints below).

Checkpoint: 2026-06-14 (DESIGN DOC: two extensibility seams ? **PR #116 OPEN, owner-gated, do NOT merge**; docs-only, branch `docs/analyzers-and-reconcilers` off master `df7d5df`). Produced `docs/ANALYZERS_AND_RECONCILERS.md` (367 lines, a peer to EXPERIMENT_HARNESS + MULTI_SIGNAL_FUSION_PLAN) + a 1-line `docs/README.md` cross-link. **Spine (owner directive): NO special-casing in the decision path** ? every analyzer/reconciler emits a SIGNAL (value+confidence); the SCORING MODEL learns to weight it (already true in miniature: `experiment.predict`/`_gate_mask` treat `net_crossings`/`players_in_court` as feature columns). **Seam 1 = injectable `WindowAnalyzer`s (forward)**: a registry (mirrors `segmenter_registry`) of pure per-window cues over the already-produced per-frame-aligned signals (WASB trajectory + `PersonDetector.detections_per_frame` + `rally_gate` net); each emits an `AnalyzerResult` = confidence-scored features ? `candidate_segments.stats`, events ? `events` table (buffer-on-candidate, flush-on-confirm), optional boundary hints; worked example = a **service-fault** analyzer (shuttle halts in net region ? `service_fault` signal + `net_hit` event + END hint); injects at the FusionSegmenter `_enrich_candidate_stats` hook, flag-gated default OFF. **Seam 2 = reconciliation pass (backward)**: a report-first "linter for rally decisions" ? `ConsistencyRule`s (trailing-dead-time?truncate *(owner's misfire fix)*, no-net-crossing?drop, over-merged?split, boundary-slack?tighten) flag decision-vs-evidence conflicts; **report-first/apply-optional, NEVER silent**; expressible as a harness `Variant` transform so LOVO-measurable on the 6 golden videos BEFORE auto-apply; idempotent, fixed precedence. ? **DESIGNED AGAINST the IN-FLIGHT fusion-P2 WIP** (uncommitted in the shared tree: `fusion_hybrid.py`, `fusion_features.py`, `FusionConfig.cue_flags`, `person_detector.detections_per_frame`, `docs/QUALITY_ITERATIONS.md`) ? that WIP was kept **byte-intact** (I isolated my single README line via git plumbing; tree is CRLF / autocrlf=true, restored owner README to exact session-start bytes 3841B). **Merge-order:** the doc cross-links QUALITY_ITERATIONS.md + the fusion hook, neither on master yet ? **merge #116 after (or with) fusion-P2** (noted in the PR body). ?? NEXT: owner reviews #116; implementation is the phased A1?A3 / R1?R3 plan in the doc (each ONE PR off master, default OFF, promoted only on measured LOVO lift) ? all owner-gated, none started. Ties [[candidate-segments-finetuning]] + the harness ([[working-agreement-merging]]).

Checkpoint: 2026-06-13 (RALLY-QUALITY TRACK: HARNESS + PERSON-CUE EXTRACTION SHIPPED ? **PR #114 + #115 MERGED**, master `df7d5df`; suite 421?441 green; 0 open PRs). **PR #115 (fusion P1):** extracted the torchvision person detector + ByteTrack + velocity windowing out of `yolo_hybrid` into `backend/pipeline/detectors/person_detector.py` (`PersonDetector`, a reusable cue producer); `yolo_hybrid` now ORCHESTRATES only (chunk/prefetch/AI-handoff/DB) and delegates Stage-1 to `self.person`. **No behavior change** ? proven by the unchanged `process_video` output tests (mock seam moved to `self.person`); pure detection unit tests ? new `tests/test_person_detector.py`. person_detector is mypy-CLEAN (`self.model: Any`) so it needed NO quarantine; the torchvision code that put yolo_hybrid in quarantine left it, but yolo_hybrid STAYS quarantined for PRE-EXISTING orchestration debt (implicit-Optional process_video defaults, Success|Failure ABC-override mismatch, a Future[...] annotation ? flagged in mypy.ini to ratchet separately). ruff+mypy clean, CI 4/4. Docs synced (fusion-plan P1?DONE, CODE_MAP person_detector entry + yolo_hybrid-as-orchestration, NEXT_STEPS item 2 ticked). ?? **NEXT (owner-steered):** **PR 3 = fusion P2.** The cheap, highest-confidence, PURE-LOGIC piece = **wire Gate A (`min_crossings`) into the live trajectory path** (`trajectory_hybrid`/`wasb_hybrid`) + add the knob to `IndexingConfig` ? *fixes the owner's held-shuttle false positive*, default OFF, immediately A/B-able via the harness on the 6 golden trajectories. Gate B (court-mask + player-count) is **owner-gated** on the court-mask-source decision (MULTI_SIGNAL_FUSION_PLAN §9.1: manual polygon vs auto homography ? owner). **P3 learned fusion + the SxS-on-new-videos RUN need GPU/owner-side** (persist person features on real video ? train on the 6 golden ? `compare_unlabeled(CONTROL, +person)` on new footage). Two PRs this session within [[working-agreement-merging]] self-merge authority.

Checkpoint: 2026-06-13 (EXPERIMENT HARNESS P1 SHIPPED ? **PR #114 MERGED**, master `ecee786`; suite 421?440 green). Picked up the rally-quality track (the desktop-agent pick-up `NEXT_STEPS` flagged; both design docs `EXPERIMENT_HARNESS.md` + `MULTI_SIGNAL_FUSION_PLAN.md` arrived via the session-start `git pull` 15e4e0e?9231af2, PR #112). Built `backend/eval/experiment.py` = the thin variant/experiment layer from `EXPERIMENT_HARNESS.md` P1 (the audit's "~80% already exists" HELD ? verified file:line; pure glue over metrics/calibration/classifier/training/regression.gt_hash/gemini_refine.merge_overlaps/query_candidate_segments, **NO new scoring math**): `Variant` (JSON recipe: segmenter+config_overrides+cue_flags+Gate-A `min_crossings`+Gate-B `min_players`+learned classifier; `spec_hash`; pinned `CONTROL`), `compare` (labeled A/B, **LOVO-honest** train-on-others for learned variants, B?A deltas + `label_set_hash`), **`compare_unlabeled`** (the owner's **SxS-on-NEW-videos** ask ? agreed/a_only/b_only via match_intervals, no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), `load_video_candidates` (DB bridge). 19 tests incl. the load-bearing **no-regression invariant** (all cues OFF ? byte-identical to CONTROL); ruff+mypy clean; CI 4/4. Docs synced (EXPERIMENT_HARNESS P1?DONE, CODE_MAP entry + stale regression-baseline-path fix, NEXT_STEPS item 3 ticked). Owner answered ramp-up Qs: **corpus = all 6** golden videos (he said "5"; LargeTestVideo/TestLargeVideo may be cuts of one match ? kept all 6 per the manifest); **merge authority = self-merge on green then continue to PR 2** (within [[working-agreement-merging]]). ?? NEXT (this session): **PR 2 = fusion P1** person/vision-cue extraction (lift torchvision+ByteTrack+velocity out of `yolo_hybrid` ? `backend/pipeline/detectors/person_detector.py` producer; yolo_hybrid consumes it, no behavior change ? regression-safe). Then P3 learned fusion trains on the 6 golden videos + `compare_unlabeled(CONTROL,+person)` SxS on new footage (GPU/owner-side). Tracks [[candidate-segments-finetuning]] + [[collector-indexer-bridge]].

Checkpoint: 2026-06-13 (WORKING-TREE CLEANUP ? master `15e4e0e`, tree now CLEAN). Cleaned up the leftover uncommitted bits (the cache-hygiene CODE was already MERGED as #109/#110 per the checkpoint below ? nothing to rescue there): (1) `config.json` GPU-run knobs **reverted to committed defaults** (was wasb_hybrid/7200/8 ? gemini/1800/1; values stay recorded in the cache-hygiene checkpoint below ? re-apply `default_segmenter: wasb_hybrid` in one line if that's the intended new default; user didn't answer keep-vs-revert, took the safe recoverable path). (2) stray root `arial.ttf` (unused ? shipped watermark font is committed `backend/assets/fonts/RobotoSlab-Regular.ttf` from #103) ? moved to `scratch/`. (3) genuine untracked draft `docs/STORAGE_SHARING_MODEL.md` (190-line storage-anchored monetization model, owner co-design pending) ? branched + **PR #111 OPEN (owner-gated, docs-only ? do NOT auto-merge**; supersedes COMMERCIALIZATION.md framing if ratified ? ADR then). Master tree now shows ONLY intentional carryovers (untracked `scratch/` handoffs+repro scripts, `artifacts/` Gen-0 bundle) ? could be gitignored for a pristine `git status` if desired (offered, not done). Reminder: `python -m tools.cleanup_caches --apply` reclaims the 96.5 GB the SessionStart hook flagged.

Checkpoint: 2026-06-13 (CACHE HYGIENE shipped ? **PR #109 + docs #110 MERGED**, master `15e4e0e`; + the ?5 reel + manual cleanup). **#110** = diligence docs follow-up (README §Cache Hygiene + knobs + tools/ line; CODE_MAP detector self-clean, tools/cleanup_caches.py entry, common-task rows; min_shot_count API-parity note) ? docs-only, CI 4/4 green. Both repos now have **0 open PRs**. After the GX010125 E2E run left ~163 GB in WSL `~/clips`, built a full cache-hygiene system (all on master): (1) **self-cleaning runners** ? `indexing.{wasb,tracknet}.keep_frames` (default false) deletes the frame cache + staged video after a SUCCESSFUL run; on failure keeps them for resume; (2) **tools/cleanup_caches.py** dry-run-first ops tool (`--apply`/`--keep-video`/`--older-than`/`--include-wasbcache`/`--summary`); (3) **disk guard** `min_free_gb`; (4) **SessionStart nudge hook** in `.claude/settings.local.json` (local, gitignored). Suite 409?**421 green**, ruff+mypy clean. Policy saved ? [[cache-hygiene-policy]]. ? WSL-scan footgun: the owner's login shell mangles `for e in *` loops (loop var empty) ? use du/find arg-globbing. Also this session (earlier): merged **#107** (API min_shot_count) + **#108** (watermark Windows colon fix); ran GX010125 E2E on GPU ? **GX010125_highlights.mp4** (11 rallies >7 shots) + **GX010125_highlights_min5.mp4** (24 rallies ?5, watermark now works ? validates #108) via a stitch-only reuse script (no re-inference); confirmed **Gemini got 1080p** (4K only at final stitch); delivered an execution flowchart. Manual WSL cleanup freed ~68 GB (kept GX010125's 93 GB frames + proxy per owner choice). ? LOCAL uncommitted `config.json`: default_segmenter=wasb_hybrid / min_shot_count=8 / wasb.timeout_sec=7200 from the run. See [[gotcha-long-runs-gpu-wsl]].

Checkpoint: 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-up session; no new code from me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**. Reconciled remote vs my last session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh): **#101** configurable branding watermark · **#102** thread config stitching paddings into live VideoCompiler sites · **#103** bundle Apache-2.0 default watermark font + fallback log · **#104** C13 field kit (flyer/playbook/answer-sheet/signals) · **#105** i18n design plan (Kannada-first) · **#107** enforce `stitching.min_shot_count` on /api/compile · **#108** escape Windows

drive-letter colon in ffmpeg filtergraph (fixes #103 watermark on Windows). **OPEN: #106** docs/video-locality-model
(awaiting owner). ? **UNCOMMITTED WIP IN TREE (sibling/owner ? DO NOT CLOBBER):** a WSL **cache-hygiene** slice in
progress, built on my `WslCommandMixin` ? `base.py` (`_wsl_free_gb`/`_wsl_rm_rf`), `wasb_runner.py`+`tracknet_runner.py`
(`keep_frames`/`min_free_gb` config + pre-run disk guard + post-success cache delete), `models.py` (new fields),
`config.json` (the GPU-run knobs: default_segmenter=wasb_hybrid, min_shot_count=8, wasb.timeout_sec=7200), untracked
`arial.ttf` (watermark font) + `docs/STORAGE_SHARING_MODEL.md`. Not yet PR'd by whoever owns it. [[session-handoff]]
rewritten to this state.

Checkpoint: 2026-06-13 (FIRST FULL E2E GPU RUN on a real 4K GoPro match + 2 merged fixes).
Owner asked to run the indexer E2E on the GPU over `F:\GoPro Hero Black 12\KhelSutra\Badminton 12
June 2026\GX010125.MP4` (12.8-min 4K60, 11.5GB), highlights for rallies with >7 shots, config-driven.
- **DELIVERED:** `output\GX010125_highlights.mp4` ? **11 rallies (shot_count 8?16) of 52 detected**;
  true 4K 3840×2160 h264 59.94fps, 2:23, 1.15GB; stitched from the 4K ORIGINAL (proxy only for indexing).
  wasb_hybrid: 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed rallies. min_shot_count=8
  correctly excluded a 7-shot rally (">7" semantics). Gemini ran clean (no 503s); ~47-min wall clock.
- **2 PRs merged on green CI** (owner's [[working-agreement-merging]] standing authorization):
  #107 `/api/compile` now enforces `stitching.min_shot_count` (parity w/ run_process_and_stitch);
  #108 stitcher `_ff_quote` escapes the Windows drive-letter colon ? fixes the on-by-default
  bundled-font watermark (PR #103) silently failing on Windows (`fontfile='C:/...'` broke ffmpeg
  drawtext parsing; also fixed absolute logo `movie=` paths). Both with regression tests; suite 409 green.
- **The hard-won run recipe is in [[gotcha-long-runs-gpu-wsl]]:** Modern Standby killed the first 2
  overnight attempts (keep-awake fix); cv2 PNG extraction (~38min) is the real bottleneck not the GPU
  (ffmpeg pre-extract, count-exact for CFR ? skips it); 1080p-proxy-index / 4K-stitch; per-phase
  observability (cache manifest for WASB, handoff dir for Gemini). Reusable scripts in `scratch/`
  (run_gx010125.py, keep_awake.ps1, probe_*.py).
- ? LOCAL UNCOMMITTED `config.json`: default_segmenter=wasb_hybrid, min_shot_count=8, wasb.timeout_sec=7200
  (set for this run; revert to gemini/1/1800 if not wanted). Watermark fix NOT applied to the delivered
  reel (it ran before #108); a re-stitch would hit the DB segment cache + now-working watermark (~5min).

Checkpoint: 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; **PR #104 updated `9c84fea` +
PR #106 OPEN `85abe26`, both CI green, both AWAITING OWNER**). Owner spec: (1) remove ALL associate-pay from the
field kit ("something else in mind") ? flyer/playbook now "Engagement terms: discussed individually"; honesty
mechanics reworded ("judged on honest capture, not positive answers"); KEPT: WTP research questions (study subject)
+ the ACADEMY's Phase-2 session fee (different party ? flagged, owner may strip too). (2) Storage/upload questions
answered from code: uploads ARE stored in full (`security.upload_dir`, NO retention

policy ? BACKLOG); 10GB CANNOT
447   upload (max_upload_mb=4096 rejects at init; ?90MB sequential parts; hours on Indian uplinks). (3) NEW
448   `docs/VIDEO_LOCALITY_MODEL.md` (#106): central insight = timestamps are the product, video is dead weight (only the
449   compiler touches original bytes, cuttable where the original lives); locality ladder L0?L4 (L2 proxy-upload reuses
450   the collector ADR-002 frame-identical contract; L1-direct = client?Gemini, zero bytes through us; L0 = owned-model

6. user (2026-06-15T14:57:59.839Z)

docs\EXECUTION_POLICY.md:102: watermark (or skip the prune on park) so a resume does not redo confirmed work.
docs\FIELD_VISIT_PLAYBOOK.md:1:# Field Visit Playbook ? academy conversations (KhelSutra)
docs\FIELD_VISIT_PLAYBOOK.md:17:KhelSutra takes ordinary practice/match video from a badminton academy and finds the rallies automatically, builds
docs\FIELD_VISIT_PLAYBOOK.md:102:record 1?2 sessions ? then we process the footage through KhelSutra and **you return to present the academy its own
docs\HOSTING_PLAN.md:1:# Hosting Plan ? khelsutra.guru
docs\HOSTING_PLAN.md:27:   ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
docs\HOSTING_PLAN.md:28:   ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on the box)
docs\HOSTING_PLAN.md:49:browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
docs\HOSTING_PLAN.md:50:   ??? api.khelsutra.guru ??? Cloud Run (FastAPI control plane, scale-to-zero)
docs\HOSTING_PLAN.md:80:1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan. Cloudflare
docs\HOSTING_PLAN.md:82:2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ?
docs\HOSTING_PLAN.md:94:| **GitHub repo** `khelsutra` (private) | Pages connects once (dashboard, you; build root = `web/`) ? every push auto-deploys | I write the SPA in your Projects folder; commits pushed = deployed |
docs\HOSTING_PLAN.md:96:| **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin, Storage admin, Cloud SQL admin) | Control-plane deploys |
docs\HOSTING_PLAN.md:104:Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ? rate
docs\archives\decisions\DECISIONS.md:471:## ADR-009: KhelSutra Guru ? product identity, web-UI repo topology, local-first delivery
docs\archives\decisions\DECISIONS.md:477:1. **Product name: KhelSutra Guru**; domain **khelsutra.guru** (GoDaddy receipt: 3-yr
docs\archives\decisions\DECISIONS.md:479:2. **The web product lives in a separate private repo `khelsutra`** ("the heart and brain"):
docs\archives\decisions\DECISIONS.md:493: product work. Named `khelsutra` (not `khelsutra-ui`) so the beta control plane doesn't force a
docs\archives\decisions\DECISIONS.md:498: failure mode is documented in ADR-008's rationale) with pointer READMEs in `khelsutra`.
docs\archives\decisions\DECISIONS.md:504:- `khelsutra` is all-rights-reserved (no OSS license file); secrets never committed; Pages build
docs\archives\decisions\DECISIONS.md:509: `khelsutra` git init/commit/push run Windows-side (owner or Claude Code).
docs\CODE_MAP.md:7:[Omitted long matching line]
docs\CODE_MAP.md:186:[Omitted long matching line]
docs\CODE_MAP.md:211:[Omitted long matching line]
docs\CODE_MAP.md:291:- **`add_video` is an UPSERT** (`INSERT ... ON CONFLICT(id) DO UPDATE`) that mutates the row in place ? re-ingesting a `video_id` does NOT cascade-delete its segments (the old `INSERT OR REPLACE` + FK-cascade wipe is gone). Segment clearing is now an explicit destroy-AFTER-confirm via `segment_watermarks`/`prune_segments` (or `clear_video_data`). `_get_connection` MUST still re-issue `PRAGMA foreign_keys=ON` per connection (CASCADE/SET NULL still apply on an explicit delete).
docs\I18N_PLAN.md:31:in a **separate private repo `avidullu/khelsutra`**. The design below is therefore deliberately
docs\I18N_PLAN.md:111:The bundled UI/watermark fonts (Outfit, Roboto Slab) **do not cover

Kannada or Devanagari**.
docs\I18N_PLAN.md:113:with the watermark font: OFL is not in the strict **MIT/Apache/BSD** code/data/weights guardrail.
docs\I18N_PLAN.md:118:Note the **video watermark already takes a `font_path`** (`WatermarkConfig.font_path`,
docs\I18N_PLAN.md:119:`backend/pipeline/stitcher/compiler.py`), so stamping a Kannada/Hindi watermark is a

[Showing results with pagination = limit: 30]

7. assistant (2026-06-15T14:58:03.731Z)

The current-state confirms the ground truth. Let me check the existing field-associate docs to ensure this overview complements (not duplicates) them, and verify the watermark text exactly.

8. user (2026-06-15T14:58:04.694Z)

Field Visit Playbook ? academy conversations (KhelSutra)

> **Recruiting?** Hand candidates [`PRODUCT_FIELD_ASSOCIATE_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
> ? that's the role description. **This document is the working playbook the hired
> associate gets on day one:** the conversation script, the ground rules, and the
> Phase-2 (recording / show-back) procedure.

Duration: ~3?4 weeks, part-time (flexible hours, mostly afternoons/evenings when academies are active)
Location: Bengaluru ? in-person visits to badminton academies
Compensation: ? ___ per completed academy visit + travel reimbursement; Phase-2 sessions paid extra ? ? ___ per session *(rates filled at hire; same numbers as the recruiting flyer)*
Reports to: Avi (phone/WhatsApp: ___)

The gig in one paragraph

KhelSutra takes ordinary practice/match video from a badminton academy and finds the rallies automatically, builds
highlight reels, and generates a simple summary (rallies found, actual play time); we're building a private web app
that a few invited academies will be able to try. We need honest, on-the-ground feedback from real academies ? **how
they work with video, whether this actually helps them, and what would make it genuinely useful**. **Your job: visit
10?15 good academies, have a friendly 20?30 minute conversation with the owner/head coach (and, with permission, a
few adult (18+) students) using the script below, and write down what they actually say.** No selling. No tech knowledge needed. For interested academies there's a Phase-2 follow-up
(a recording session ? then you return and show them THEIR OWN footage as highlights) with additional compensation.

Phase 1 ? The conversation (what you actually do)

Who to talk to: the academy owner or head coach ? the person who decides what the academy spends money on.
If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

Opening (keep it this honest):
> "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice

> video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to
> understand how you work with video today. Could I ask you a few questions? About 15?20 minutes, and we're not
> selling anything today."

The questions (ask in this order; the starred ones matter most):

1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?
2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,
   tournament days, CCTV) ? roughly how much?
3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?
4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, Stupa, BadPro+, just
   YouTube/phone.) What do you like / what's missing in it?
5. ? **If you could hand over a full session's video and get back the rallies cut out + a highlight reel + a simple
   summary ? would you pay per match for that? What would that be worth to you, per match?** Then **stop talking and
   wait** ? let THEM name the number. If they genuinely won't after a pause and one "roughly?", you may offer the
   ladder: "?100, ?300, ?500?" **On the sheet, tick whether their number came BEFORE or AFTER you offered the
   ladder** ? it changes what the answer means. (Note their face, not just the words.)
6. Your hall has multiple courts running at once, right? Does that make video less useful today?

? NOW (and only now) show the sample reel ? a ~60-second real highlight reel we put on your phone. Say: *"this is
what it produces ? the rallies found and cut automatically."* Then shut up and watch their face. Write down their
FIRST reaction verbatim, and ask one follow-up: **"what would this need to do for it to be useful for YOUR academy?"**
(Improvement asks, in their words, go on the sheet. Showing it earlier anchors every answer above ? never lead with it.)

7. Would you let us record one session here (we pay a small fee for the time) and come back showing what the tool
   produces on YOUR court, your players? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

9. user (2026-06-15T14:58:06.379Z)

backend\assets\fonts\README.md:11:`RobotoSlab-Regular.ttf` is the **default** watermark font: `stitching.watermark.font_path`
backend\config\models.py:178:   text: str = "Generated by Khelsutra.guru"
backend\config\models.py:194:   watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
backend\pipeline\stitcher\compiler.py:10:# Bundled, commercially-clean default watermark font (Apache-2.0). Shipped so the overlay
backend\pipeline\stitcher\compiler.py:47:   on-by-default bundled-font watermark (and absolute logo paths) on Windows."""

10. assistant (2026-06-15T14:58:10.381Z)

I have a key accuracy finding: the actual default watermark text is "Generated by Khelsutra.guru" (lowercase 's', no capital S), but the draft writes "Generated by KhelSutra.guru" (capital S). This is a minor but verifiable discrepancy worth flagging. Let me confirm by reading the exact config line.

11. user (2026-06-15T14:58:11.471Z)

```
174         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
175         (under `stitching.watermark` in config.json) to turn it off. The text is fully
176         configurable. The concat stage stays stream-copy (-c copy)."""
177         enabled: bool = True
178         text: str = "Generated by Khelsutra.guru"
179         logo_path: str = ""        # optional transparent PNG overlay; "" = no logo
180         font_path: str = ""        # optional .ttf; "" = use the fontconfig `font` family below
181         font: str = "Sans"         # fontconfig family used when font_path is empty
182         font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5)  # text height as a fraction of frame height
183         opacity: float = Field(default=0.85, ge=0.0, le=1.0)
```

12. assistant (2026-06-15T14:58:14.852Z)

I have enough verified ground truth. The watermark text discrepancy ("KhelSutra.guru" vs the actual default "Khelsutra.guru"), the field-playbook complementarity, and all the ground-truth claims are confirmed. Let me now produce the structured review.

13. user (2026-06-15T14:59:06.943Z)

Structured output provided successfully

14. assistant (2026-06-15T14:59:15.539Z)

I've completed the review of the KhelSutra product-overview draft against the repo's ground truth. Findings are in the structured output. Key points:

Verified against source:
- The watermark text discrepancy is real and load-bearing: `backend/config/models.py:178` ships `"Generated by Khelsutra.guru"` (lowercase 's'), but the draft writes `"KhelSutra.guru"` (capital S) at lines 11 and 45.
- All guardrails (opt-in, minors-never-used, local-first, pricing-not-set) and the TODAY-vs-ROADMAP split match the current-state memory and the existing field docs. The doc correctly complements `FIELD_VISIT_PLAYBOOK.md` / `PRODUCT_FIELD_ASSOCIATE_FLYER.md` rather than duplicating them.

Two must-fixes: the watermark text mismatch, and the "finds **every** rally" overpromise (line 10) that the doc's own multi-court caveat contradicts.

Biggest gap (should-fix): for the alpha-USER half of the audience, the doc explains *what* the tool does but never *how* to use it (no upload/async-job/where-the-reel-appears walkthrough) and never tells users *what feedback* the team wants ? both are explicit goals in the brief.

The document at `C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md` is read-only per the task; I made no edits.